

MC-GPU Phantom Pipeline

A set of 158 Monte-Carlo voxel phantoms built on one fixed grid from ICRP mesh anatomy and patient head CT, with the acquisition geometry they are placed against and the statistics of the set as it stands.

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158 PHANTOM VOLUMES	110 DISTINCT ANATOMIES	0.3 mm ISOTROPIC VOXEL
3.80 G VOXELS PER VOLUME	8 MATERIAL LABELS	98 VOLUMES PUBLISHED

CONTENTS

- 01

The phantom set
- 02

Acquisition geometry
- 03

Anatomical sources
- 04

Quality control
- 05

Coverage & caveats

01 The phantom set

Every phantom, whatever its anatomical origin, occupies the same fixed grid with the same eight material labels. A simulation campaign can therefore substitute one phantom for another without altering the beam definition.

ANATOMICAL SOURCE	CONTENT	ANATOMIES	VOLUMES	STATUS
ICRP mesh series	newborn, 1, 5, 10, 15 y and adult, both sexes	12	12	published
ICRP, support variants	5 sampled patient-support positions per phantom	12	60	archived
GradientHealth	full cranium, masks segmented by hand	10	10	published
CQ500, full cranium	8 native series + 2 recomposed from partial acquisitions	10	10	published
CQ500, partial cranium	vertex outside the field, upper dentition in it	66	66	published
Total		110	158	98 published

Support variants share their anatomy with the base ICRP volumes and differ only in the position of the patient support, so they add volumes without adding anatomies.

Volume specification

EXTENT 640.2 × 640.2 × 250.2 mm	VOXEL 0.3 mm isotropic	MATRIX 2134 × 2134 × 834
VOXEL COUNT 3.80×10^9	SIZE ON DISK 3.80 GB, 8-bit unsigned	ORDER Z-major, X fastest
ORIGIN isocenter, at the volume centre	AXES +Y posterior, +Z cranial	COMPRESSED 2.7 – 5.5 MB per volume

Material labels

Labels 0–4 are assigned by thresholding the Hounsfield volume; labels 5, 10 and 11 are geometric bodies written after classification and are therefore independent of the thresholds and of any noise treatment.

LABEL	MATERIAL	ASSIGNMENT [HU]	DENSITY [G/CM ³]
0	Air	< -500	0.0012
1	Adipose tissue	-500 ... -50	0.95
2	Soft tissue	-50 ... 200	1.05
3	Bone, spongiosa	200 ... 800	1.25
4	Bone, cortical	> 800	1.5 – 1.9
5	Implant, titanium	geometric	4.54
10	Support shell, carbon	geometric	2.00
11	Support core, PU foam	geometric	0.021
Tissue densities for the mesh phantoms are the phantom's own age-specific values; cortical bone spans roughly 1.5 g/cm ³ in the newborn to 1.90 g/cm ³ in the adult. Patient-derived phantoms use the adult female reference set throughout.			

HOUNSFIELD VALUES ARE DISCRETE, BY CONSTRUCTION

A mesh phantom assigns one attenuation coefficient per tissue, so its Hounsfield volume is a set of discrete values rather than a continuous distribution. At 60 keV monoenergetic, cortical bone reaches 1500–4700 HU against 1000–2000 HU in clinical polychromatic CT, with an empty interval between spongiosa and cortical bone. Label assignment is unaffected, but mesh- and patient-derived phantoms are not comparable at the Hounsfield level.

02 Acquisition geometry

One beam definition serves the whole set. Because the reconstructed field of view is considerably shorter than the output volume, where the anatomy sits inside that volume is a defining property of the dataset rather than a detail of its production.

SOURCE-AXIS DISTANCE 578.0 mm	SOURCE-DETECTOR DISTANCE 987.0 mm	DETECTOR 293.4 × 293.4 mm, CsI 0.6 mm
SIMULATED MATRIX 512 × 512	TUBE VOLTAGES 80 kV, 120 kV	FILTRATION Al 340 μm · Al 100 μm + brass 60 μm

Reconstructed field of view

By similar triangles the reconstructed FOV at isocenter is only ±85.9 mm tall, against a 250 mm output volume. Anatomy placed against the top face of the volume rather than against the beam therefore left approximately **64 mm of head outside the reconstructed field** and invisible to every projection.

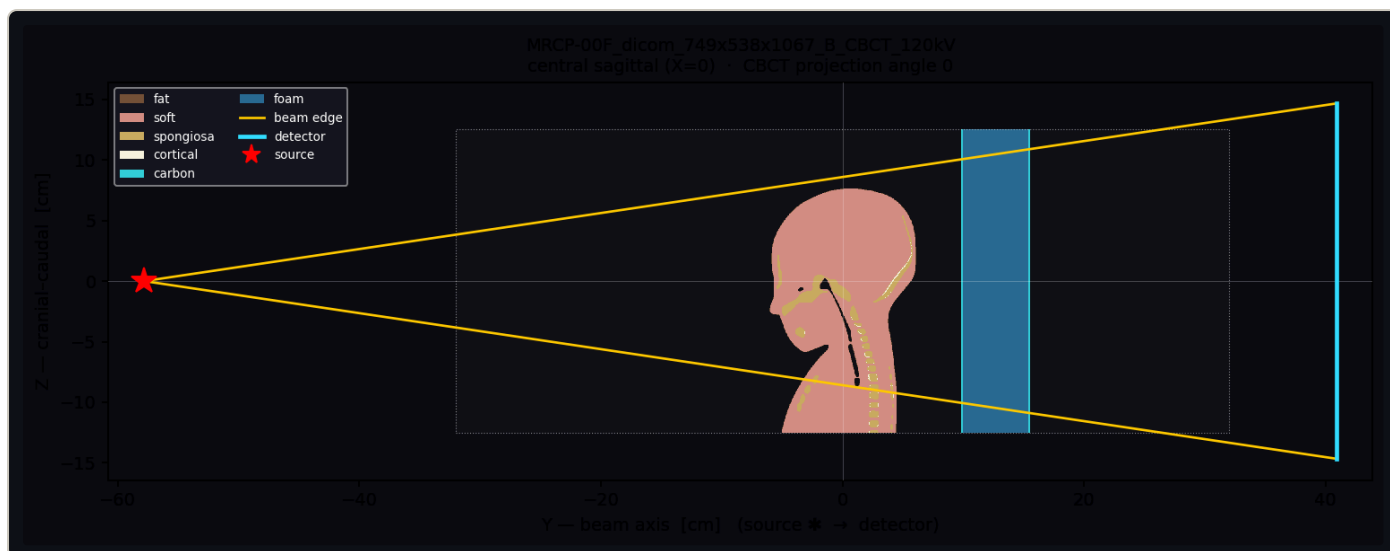
$$\begin{aligned}\text{FOV half-height} &= (\text{detector height} / 2) \times (\text{SAD} / \text{SDD}) \\ &= 146.7 \text{ mm} \times (578.0 / 987.0) = 85.909 \text{ mm}\end{aligned}$$

The placement rule adopted for the whole set anchors the detected cranial vertex at **75.909 mm** above isocenter — 10 mm of air below the top edge of the reconstructed field — and centres the transverse position on the bounding box of the uppermost 100 mm of anatomy, so the head is centred as in a real head examination rather than being pulled off-axis by the shoulders.

Patient support

The support cross-section was reconstructed from real CBCT projection data rather than from a drawing: a rounded trapezoid 440 mm wide at the top and 396 mm at the base, 57 mm high, formed from a 1.5 mm carbon wall around a polyurethane foam core, with a 4 mm chamfer and 5 mm corner radii.

It is seated automatically behind the measured posterior surface of the head with a 2 mm clearance. Across the 66-case patient batch the resulting support position ranged from **119.3 to 150.6 mm** posterior to isocenter, tracking head depth. Sampled displacements around that seat generate the support-position variants, each verified to clear all tissue voxels by at least the nominal 2 mm before being written.



Newborn mesh phantom, support variant B. Central-sagittal section of the label volume with the source (red), beam edges (amber) and detector (cyan) at projection angle 0. The cranial vertex sits inside the field with the specified air gap; the support shell — carbon wall and foam core — is at one of the sampled positions. Head proportions are those of the newborn reference phantom.

03 Anatomical sources

ICRP mesh series

Twelve tetrahedral reference phantoms covering newborn to adult in both sexes. Attenuation is computed per tissue at 60 keV and converted to Hounsfield units; the adult female phantom alone resolves **227 distinct tissue materials** before they are pooled into the five-tissue scheme. The captured anatomy spans a fixed 320.1 mm cranio-caudal window from the vertex downward, so the transverse extents below measure body size across the series.

PHANTOM	MATRIX	LATERAL [MM]	A-P [MM]	CRANIO-CAUDAL [MM]
Newborn, F	749 × 538 × 1067	224.7	161.4	320.1
Newborn, M	749 × 538 × 1067	224.7	161.4	320.1
1 year, F	971 × 686 × 1067	291.3	205.8	320.1
1 year, M	950 × 685 × 1067	285.0	205.5	320.1
5 years, F	1172 × 790 × 1067	351.6	237.0	320.1
5 years, M	1179 × 790 × 1067	353.7	237.0	320.1
10 years, F	1328 × 815 × 1067	398.4	244.5	320.1
10 years, M	1317 × 828 × 1067	395.1	248.4	320.1
15 years, F	1316 × 812 × 1067	394.8	243.6	320.1
15 years, M	1444 × 834 × 1067	433.2	250.2	320.1
Adult, F	1539 × 955 × 1067	461.7	286.5	320.1
Adult, M	1552 × 970 × 1067	465.6	291.0	320.1
Captured anatomy per phantom, sampled at 0.3 mm isotropic before placement into the common output grid. Lateral extent grows by a factor of 2.07 from newborn to adult male.				

Patient computed tomography

Eighty-six head examinations from two archives: ten from GradientHealth and seventy-six from CQ500. Slice pitch is 0.625 mm nominal; twelve further CQ500 candidates were rejected for a Z sampling of 3 mm or coarser, an order of magnitude above the 0.3 mm output voxel. Two of the ten full-cranium CQ500 cases are recomposed from two partial acquisitions of the same series, so that group is eight native series plus two rebuilt volumes.

Every case carries a tissue mask removing the couch and head cradle, which would otherwise enter the phantom as tissue. Provenance differs by archive: the ten GradientHealth cases were segmented by hand in 3D Slicer; the CQ500 cases use threshold-and-largest-component masks from the source curation, of which 22 in the partial-cranium group were re-segmented because they had included the cradle or the rail.

Gantry tilt is accounted for. A quarter of the CQ500 partial-cranium series were acquired with a tilted gantry, between 4° and 24°. The acquisition's direction cosines are resolved during resampling, so tilted series are reproduced without shear — tissue extents match the geometry declared in the series headers to better than half a millimetre on every case.

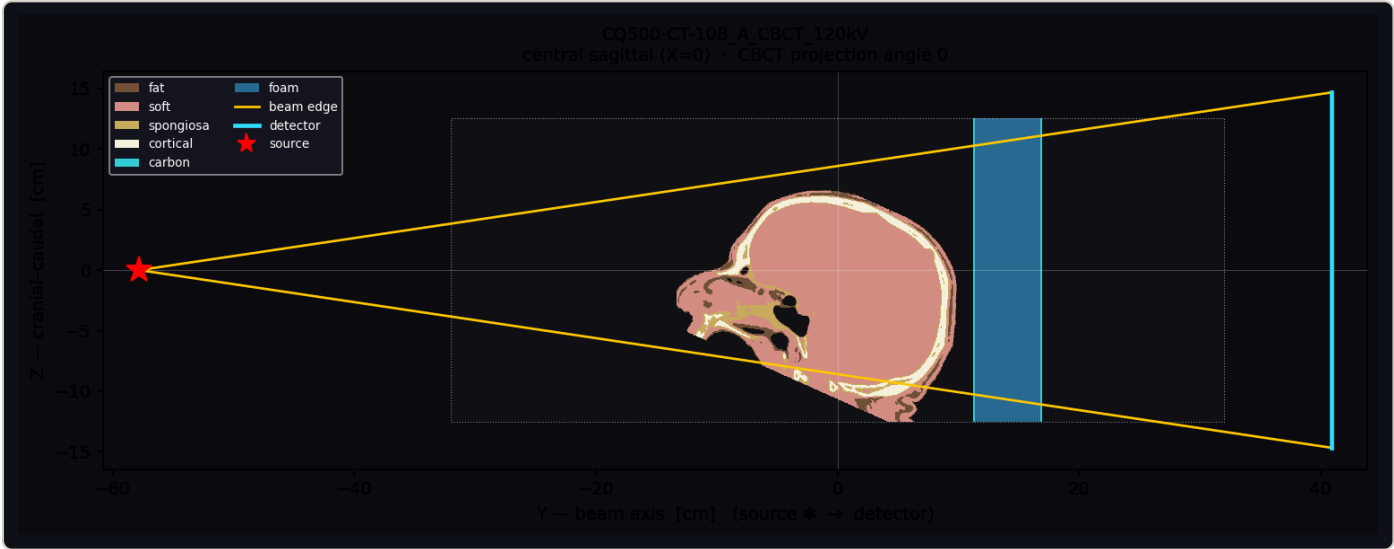
Field truncation is inherent to clinical data. Acquired to answer a clinical question rather than for dosimetry, a proportion of these series are truncated, most often at the vertex. Every case is inspected before it enters the set; those retained have truncation that is minimal and does not affect the simulated projections.

04 Quality control

Placement errors do not raise exceptions: a coordinate-frame inconsistency can put the anatomy entirely outside the beam while every reported offset remains plausible. Each volume therefore carries a rendered sagittal section with the beam overlaid, and each batch is additionally screened on distribution statistics.

QUANTITY	EXPECTED	OBSERVED ACROSS 66 CASES
Output matrix	2134 × 2134 × 834	identical, 66 / 66
Vertex anchor above isocenter	75.909 mm	identical, 66 / 66
Tissue fraction of the volume	non-zero, narrow	2.29 – 3.99%
Head A-P half-depth	anatomical range	88.8 – 120.1 mm
Support seat, posterior to isocenter	tracks head depth	119.3 – 150.6 mm

A vertex anchor identical to four decimal places across every case confirms that field-of-view anchoring engaged in all of them; a tissue fraction near zero would indicate anatomy placed outside the volume. Sections were then reviewed individually for the flagged cases and for the numerical extremes. 327 sections have been rendered and reviewed across the set.



Patient-derived phantom, quality-control section. A partial-cranium case from the CQ500 partial-cranium group, shown in the same terms as the figure above. The vertex is anchored inside the reconstructed field and the support is seated behind the occiput. The caudal end reaches the floor of the volume: the 250 mm axial window is fully occupied by the largest heads, which is a property of the fixed grid rather than of this case.

Noise treatment

Hard thresholds applied to clinical CT convert quantum noise into isolated misclassified voxels, each of which becomes a spurious material boundary for the transport calculation. A 3×3×3 median filter followed by removal of connected components smaller than 20 voxels — applied to air, spongiosa and cortical bone only, and always preserving the largest component of each — removes **99.5%** of sub-20-voxel speckle and reclassifies **3.2%** of voxels. Both stages act on the Hounsfield volume before resampling, and before the anatomy is located, so vertex detection is not misled by noise.

Production and storage

BUILD TIME 1.9 min per volume	LARGEST SINGLE BATCH 66 volumes, 2 h, 66 / 66 succeeded	UNCOMPRESSED 3.80 GB per volume, 600 GB total
COMPRESSED 2.7 – 5.5 MB per volume	COMPRESSION RATIO 930 : 1 median	WORKING DISK RECLAIMED 332 GiB
SAGITTAL SECTIONS RENDERED 327	SIMULATION INPUTS 382, at 80 and 120 kV	RETENTION archive + parameters; volumes regenerable

A volume is deleted from the working disk only once two independent copies are proven: the local archive must match the checksum reported for the uploaded object, and the archive must pass its own integrity test. Deletion is never gated on an exit status.

05 Coverage staged for expansion, and caveats

Two further archives have been screened and converted to a readable form. Nothing has been built from them yet; the counts below are what is available to build.

COLLECTION	GROUP	CASES AVAILABLE	SAMPLING
Whole-body CT 117 organ masks per case	Head, near-full cranium	194	1.5 mm isotropic
	Knee	22	
	Lower leg	5	
	Foot and ankle	5	
Lower-limb skeletal	Pelvis to toes, 21 bone labels	30	0.25 – 0.7 mm
Total		256	
The whole-body collection was screened across 1228 cases and the 226 selected are filed by most distal anatomy, so each appears once. Thighs were excluded deliberately — almost all are incidental femur in abdominal scans rather than limb studies. The lower-limb collection is inventoried but not yet converted; 20 of its 30 cases are stored inverted and need their supplied transform applied first.			

Caveats on the present set

OBSERVATION	CONSEQUENCE	STATUS
Support seated too far posterior on patient cases	Head depth is measured before the tissue mask is applied, so the couch and head cradle count toward it. The support sits 20–40 mm further back than the nominal 2 mm clearance on the CQ500 cases, and about 80 mm on the GradientHealth ones. Correctable per case by a negative displacement.	quantified
Caudal extent limited by the volume	The 250 mm axial window is fully occupied by the largest heads, so the most inferior slices of those cases fall at the face of the volume. The window was left unchanged to keep the set dimensionally uniform; the anchored vertex end is unaffected.	by design
Age-specific spectra not applied to mesh variants	Where the mesh phantoms are exported on the common grid directly, the adult tissue set is used for all ages. Geometry is unaffected; a small number of voxels near the spongiosa–cortical boundary may be assigned differently than an age-specific set would give.	by design

Next increments

- Build the head group of the whole-body collection — 194 cases, which roughly triples the patient-derived part of the set.
- Measure head depth on the masked volume, removing the 20–80 mm support displacement on patient cases.
- Extend beyond the head: the staged knee, lower-leg and foot material is the first anatomy in the set outside the cranium.